

Docking–Molecular Dynamics Divergence as a Discovery Signal for African Antimalarial Natural Products

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Abstract

No study has systematically calibrated docking reliability for African antimalarial natural products, a chemical class whose rigid polycyclic frameworks and rare functional groups may challenge scoring functions. We computed docking-derived resistance-retention scores (RRS) on 136 PfDHFR and PfCRT wild-type and mutant systems for a 17-candidate cohort from 65 856 molecules derived from African natural products. The primary analysis on 12 two-target candidates produced a five-class stratification that separates predicted wild-type potency from predicted mutant retention. A 16-system single-replicate pilot on two candidates revealed that docking-RRS

and short-MD geometry metrics diverged in direction for seven of eight mutant comparisons (87.5% divergence rate). This elevated divergence rate constitutes a diagnostic signal identifying candidates whose docking ranking is likely an artifact of static pose scoring. An external 39-ligand panel confirmed that docking-score retention is protocol-dependent. The study supports computational prioritization as a hypothesis-generating tool but does not establish biological target engagement, affinity, or resistance resilience.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden[?]. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries[?], and declining efficacy of some partner drugs has been reported^{??}. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding[?], while genomic surveillance documents the co-occurrence of *Pfprt* and *Pfdhfr* resistance variants across diverse transmission settings^{???}.

African natural products represent an underexplored chemical space for antimalarial drug discovery. A systematic review identified 652 plant taxa with documented antimalarial activity, yielding 134 prioritized compounds from 39 species[?], yet fewer than 5% of African NPs have been validated computationally[?]. This gap is critical because African NPs present molecular characteristics—rigid polycyclic frameworks, macrocyclic systems, and rare functional groups—that challenge scoring functions calibrated for drug-like molecules^{??}. Under the single-drug–multi-target (SDMT) paradigm, simultaneous loss of activity at several relevant targets raises the mutational barrier to treatment failure^{??}, and natural products are attractive starting points because their stereochemical complexity supports activity across multiple protein classes^{??}. However, a list of predicted protein contacts is not a mechanism-

of-action assignment; causal network analyses require explicit perturbation models and evidence that predicted interactions propagate to a phenotype[?]. No systematic calibration of docking reliability for African antimalarial NPs exists, leaving a blind spot in the computational selection pipeline.

In a previous study, we used generative molecular methods and dual-filter consensus docking (AutoDock Vina[?] and DiffDock-L[?]) to generate a library of 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs[?]. We selected the present candidates against a panel spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and the 4GM2 entry represents PfClpR, not PfClpP, and is retained as a proteostasis-related parent-study system. This panel is deliberately a mechanistic cross-section rather than an exhaustive parasite targetome. It allows us to ask whether resistance-aware prioritization remains informative across distinct target classes, while avoiding the stronger claim that the selected compounds define a complete parasite-wide polypharmacology profile. PfCRT docking and MD were performed at physiological pH; results for this transporter are therefore conditional on that preparation and may differ under digestive-vacuole pH.

Rigid docking is efficient for ranking but treats the receptor and scoring landscape in a simplified manner. MD tests whether a docked pose remains compatible with protein flexibility, whereas endpoint methods such as MM-GBSA provide system-specific energetic summaries rather than calibrated thermodynamic observables^{??}. This distinction is especially important for resistance analysis: a docking RRS is a hypothesis about relative mutant sensitivity, not a measurement of a mutant free-energy difference. Network pharmacology provides a complementary prioritization layer, but centrality is a topological weighting heuristic rather than a causal measure of target essentiality, consistent with the growing interest in multi-target antimalarial design more broadly[?]. We therefore treat these quantities

separately: docking-derived RRS for mutant sensitivity, PNS for network-weighted target ranking, ACSI for chemical-space positioning, and targeted MD for post-docking structural scrutiny.

Workflows that combine a docking-derived retention metric, network pharmacology and short MM-GBSA endpoints are sometimes presented as joint evidence for resistance-resilient polypharmacology, but each layer measures a different quantity and has its own scope of inference. Treating them as independent evidence of mechanism, rather than as parallel ranking signals with non-overlapping failure modes, inflates the apparent epistemic reach of a single short MD pilot. We make this distinction explicit throughout.

The central question is whether the docking-MD divergence rate constitutes a reliable diagnostic signal for African antimalarial natural products. Specifically: (i) whether resistance-aware polypharmacology docking produces a discriminant prioritization that separates candidates by their predicted retention across clinically relevant mutant panels, and (ii) whether the directional divergence between docking-RRS and short-MD geometry metrics can be used as a diagnostic signal to identify candidates whose docking ranking is likely an artifact of static pose scoring. The first question addresses whether RRS-based stratification is informative enough to generate hypotheses for downstream experimental testing. The second addresses a critical gap in the literature: no study has systematically quantified the docking-MD divergence rate for African antimalarial NPs, despite evidence that scoring functions struggle with the structural complexity of natural products^{7 8}. The primary analysis used 12 candidates with eligible PfDHFR and PfCRT wild-type scores; five PfCRT-only candidates formed a coverage-limited sensitivity set. RRS was calculated from empirical Vina scores, whereas PNS and ACSI provided complementary ranking dimensions. Four parent-study wild-type complexes—chemically and numerically distinct from the 17 Set-C candidates—underwent 10 ns simulations as a methodological pose-retention check; a 16-system Set-C pilot provided the only trajectory-level comparison against docking-RRS for members of the prioritization cohort. Biological polypharmacology and resistance circumvention were not

tested.

Methods

Study design

The primary analysis is docking-score retention (RRS) for the 12 Set-C candidates that have eligible PfDHFR and PfCRT wild-type scores. RRS is the operational baseline against which the central comparison—the directional divergence between docking-RRS and short-MD structural retention—is tested. We therefore distinguish the primary docking-RRS analysis from the central divergence comparison, secondary structural-retention diagnostics, technical validation, and unobserved biological quantities. The 16-system Set-C MD pilot provides the matched-row comparison for the divergence signal; the five PfCRT-only candidates and the four-complex parent-study MD check are reported separately as secondary analyses. They address different coverage and sampling questions and are not combined with the primary RRS classes. The parent-study MD systems are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Candidate selection and filtering

Seventeen candidates were selected from a library of 65 856 molecules[?] by sequential filtering: $\text{MPO} \geq 0.70$ (19 913 primary leads retained), $\text{SYBA} > 0$ (synthesisable), $\text{SI} > 10$ (selectively antiparasitic), multi-target engagement of at least two docking targets, and scaffold diversity $\leq$ three compounds per Bemis–Murcko scaffold. The selection includes one previously named consensus lead (Ligand 87, PfDHFR). Four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were not part of the 17-candidate docking set but were used for the production-MD wild-type complexes; the 164 system is PfClpR (PDB 4GM2, not PfClpP). Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were reference controls (Table 1). Class frequencies are conditional on these

p2_cohort_workflow.pdf

Figure 1: *Study design. Docking-derived RRS, ACSI, and PNS are computed on the 17 Set-C candidates across 136 WT/mutant PfDHFR and PfCRT systems. Four parent-study complexes (non-overlapping with the Set-C list) undergo a single 10 ns production MD per system (CHARMM36m/GAFF2 or OpenFF/CHARMM36m for the parent-study and Set-C cohorts, respectively); a 16-system pilot (PP-01 and PP-02, single 10 ns replicate per mutant) is reported separately and does not define the RRS classes. None of the MD trajectories is replicated at the level recommended by the JCIM MD-reporting guidelines[?].*

filters and should not be extrapolated to the full library.

Table 1: Breakdown of the 136 docking protein–ligand systems by target. The 17 Set-C candidates were docked against 5 PfDHFR states (WT + N51I, C59R, S108N, I164L) and 3 PfCRT states (WT + K76T, K76A). Production MD was performed separately on four non-overlapping parent-study wild-type complexes (Ligands 164/PfClpR, 201/PfDHFR, 214/PfCRT, 438/PfATP4; 10 ns each) and is not shown here.

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	—	—	136	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six resistance-associated states for comparison (Table 2); target identifiers are listed in the Supporting Information (Table S14). Each state corresponds to a WHO-listed molecular marker of antimalarial drug resistance, giving the panel a clinical anchor rather than an arbitrary residue choice[?].

Table 2: Resistance mutation panel.

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	PfCRT resistance/comparator state	Not a prevalence estimate

Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation (GMQE), Ramachandran analysis (MolProbity), and C α RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0 , GMQE exceeded 0.7 , Ramachandran favored residues exceeded 95 %, and backbone RMSD was below $1.5 \text{ \AA}$. All six mutant models (PfDHFR N51I, C59R, S108N, I164L; PfCRT K76T, K76A) met these thresholds and were retained as docking receptors; a summary is given in the Supporting Information (Table S2). K76A was included as a charge-neutralising isosteric perturbation of the canonical K76T substitution, allowing the contribution of the charged Lys76 anchor to be examined separately from the threonine substitution. Model scores are not experimental structures, and the lack of mutant crystal structures or replicate ensembles remains a source of uncertainty.

Docking protocol assessment

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study[?]: AutoDock Vina 1.2.7[?] with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ[?]), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). As an independent boundary audit of these fixed boxes, P2Rank[?] pocket predictions were compared with the grid centers; the concordant PfCRT result and the frame caveats for the other receptors are presented in the Supporting Information (Figure S1).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (Table S1). On the MMV Malaria Box consensus

(Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450, $EF_{5\%} = 0.00$), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was $< 2.0 \text{ \AA}$ relative to the crystal pose; the MTX redock attempt was a clear failure ($\text{RMSD} \approx 30 \text{ \AA}$). AutoDock Vina was used for all reported Set-C scores; GNINA yielded no usable score for this panel. Docking scores are empirical scoring approximations that are sensitive to the scoring function and protocol^{???}; they should not be treated as binding free energies.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO-affinity association was null and exploratory; complete correlations are provided in the Supporting Information (Tables S8 and S9).

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information (Figure S2 and Table S3). These predictions were not experimentally validated.

Docking-derived resistance-retention score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with $25 \text{ \AA}$ dimensions, matching the audited V2 configuration files.

For target t , let $s_{\text{Vina},i,\text{WT},t}$ and $s_{\text{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\text{Vina},i,\text{WT},t}| < 5.0 \text{ kcal mol}^{-1}$ are excluded from that target’s score-retention analysis. The

target-specific resistance-retention score (RRS) is

$$\text{RRS}_{i,m,t} = \frac{|s_{\text{Vina},i,m,t}|}{|s_{\text{Vina},i,\text{WT},t}|} \times 100. \quad (1)$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or biochemical resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. To avoid letting unequal target coverage drive the main ranking, the primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least 7.0 kcal mol⁻¹.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below 7.0 kcal mol⁻¹.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below 80 %.
- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D**: no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the distribution of Vina scores in this panel and have not been calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is reported in the Supporting Information (Table S6).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (Table S4). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S7). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m[?]/GAFF2[?] at 310.15 K using GROMACS[?] . The 10-ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy[?] . We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis[?] was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

MM-GBSA endpoint estimates were obtained from production trajectories using the GB^{OBC2} model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3[?] CHARMM36-to-AMBER conversion. These are endpoint enthalpic scores, not rigorous free energies, and their limitations for ranking dissimilar complexes are well documented[?] . Predicting mutation effects via endpoint methods remains challenging, with best-case correlations around 0.44 even at 100 ns[?] . Configurational entropy is not included. Uncertainty is reported as the standard error of the mean over stored snapshots (101 for the interpretable PfCRT-214 system; 100 for Set-C pilot systems). Within-trajectory and propagated standard deviations are tabulated in the Supporting Information and are not replicate uncertainties.

The 16-system Set-C pilot used the same temperature and length, with ligand parame-

ters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks.

Results

Operational reproducibility of the docking-RRS signal

The primary docking analysis establishes the operational reproducibility of the RRS signal that is subsequently tested against the MD layer. Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (Table S5). The difference is in docking-score retention between panels, not evidence of different biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. When all 17 candidates are classified on whatever targets are eligible, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1; the five PfCRT-only compounds cannot be compared on equal footing with the two-target set. Per-compound RRS profiles across all six mutants are shown in Figure S3. Varying the wild-type eligibility threshold from 4.0 kcal mol⁻¹ to 5.0 kcal mol⁻¹ left class counts unchanged; raising retention thresholds from 70%/80% to 80%/90% increased Class D from 1 to 9 (Table S6).

The docking-RRS signal is operationally reproducible under the declared protocol: PP-01 multi-seed dispersion is ± 0.05 kcal mol⁻¹ for PfDHFR and ± 0.04 kcal mol⁻¹ for PfCRT, and PP-15 multi-seed dispersion is ± 0.03 kcal mol⁻¹. In a bounded score-perturbation audit, 77.9% of RRS classes remained unchanged under ± 1 kcal mol⁻¹ perturbation. However, this reproducibility characterizes protocol-level consistency rather than pose accuracy or biological binding fidelity.

Chemical-space and network descriptors (secondary)

The mean ACSI for the 17-candidate cohort was 0.543, with two candidates (11.8 %) exceeding 0.70 on the approved-drug/African-NP reference space; weight-sensitivity checks are in the Supporting Information (Table S4). PNS was only weakly associated with RRS on the two-target set ($\rho = -0.2098$, adjusted $p = 1.0000$); PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S7). At $n = 12$ the wide bootstrap intervals indicate that the three ranking dimensions capture distinct information rather than redundant signals.

Cross-metric associations (secondary, post-selection)

These secondary, post-selection comparisons test whether the docking-derived RRS aligns with PNS and ACSI within the same cohort; they are not independent validation of any biological mechanism.

Table 3: Cross-metric Spearman correlations on the complete two-target set ($n = 12$). Permutation p -values use 100 000 permutations; adjusted p -values use Bonferroni correction for three tests. Bootstrap intervals are percentile 95 % intervals from 10 000 resamples. At this sample size the correlations are descriptive.

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95% CI
PNS vs. RRS	H1	-0.209 8	0.514 4	1.000 0	[-0.76, 0.54]
ACSI vs. RRS	H2	-0.405 6	0.192 2	0.576 6	[-0.89, 0.29]
RRS vs. weakest WT score	H3	-0.166 1	0.603 8	1.000 0	[-0.73, 0.59]

On the two-target set ($n = 12$), all three correlation coefficients were non-significant after Bonferroni correction: PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, RRS-weakest WT score $\rho = -0.1661$ (Table 3). PNS shares an input with RRS (wild-type Vina scores), so the PNS-RRS pair is not independent. On all 17 candidates, PNS-RRS strengthened ($\rho = -0.5588$, adjusted $p = 0.0667$), but unequal target coverage limits interpretation. Bootstrap intervals are wide, as expected for $n = 12$.



Figure 2: Cross-metric Spearman correlations for Set-C candidates. The primary analysis uses the 12 complete two-target candidates; the 17-candidate matrix is shown only for comparison when target coverage is incomplete. PNS already includes wild-type docking scores, so that pair is not an independent test.

Computational prioritization intersection (exploratory)

Requiring Class A or A*, ACSI > 0.5, and above-mean PNS leaves one candidate on the two-target panel. That intersection is reported as a shortlist for experiment, not as evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort, secondary)

Four non-overlapping parent-study wild-type complexes were simulated for 10 ns to test whether docked poses remain bound on a short timescale; these compounds are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Table 4: Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Table 5: Endpoint interpretation for four parent-study complexes used for production MD. Only PfCRT-214 yields an interpretable MM-GBSA endpoint; dissociated systems are reported as N/A.

Ligand	Target	ΔG_{bind} (kcal mol ⁻¹)	SEM	Interpretation
214	PfCRT-WT	-18.25	0.40	Interpretable; ligand bound [†]
164	PfClpR-labelled cohort-WT	—	—	N/A; ligand dissociated [‡]
201	PfDHFR-WT	—	—	N/A; ligand dissociated [§]
438	PfATP4-WT	—	—	N/A; conversion corrupted*

[†]Ligand remains bound (minimum distance 3.19 Å, 78 contacts); conversion via `gmx_MMPBSA` native converter.

The quoted uncertainty is the standard error of the mean over the 101 stored snapshots (see Methods).

[‡]Ligand dissociates to 67.4 Å minimum distance with zero contacts during production; excluded from binding-mode conclusions.

[§]Ligand dissociates to 78.2 Å minimum distance with zero contacts; same caveat applies.

*Excluded: conversion artifact in the two-chain PfATP4 topology (+473 kcal mol⁻¹ van der Waals inflation); structural persistence is reported separately.

Two of four systems (PfCRT-214, PfATP4-438) retained their ligands, while the 164-



Figure 3: Stability analysis for the parent-study PfCRT–Ligand 214 complex over 10 ns production MD. (A) Protein backbone RMSD; (B) ligand heavy-atom RMSD; (C) protein–ligand contacts within 5 Å; (D) protein–ligand hydrogen bonds per frame. Dashed lines in (A) and (B) indicate stability thresholds. The system retains a bound ligand with persistent contacts and H-bonds; cohort-level metrics are in Table 4.

labelled PfClpR and PfDHFR-201 ligands dissociated completely despite stable protein conformations (Table 4). Only PfCRT-214 yielded an interpretable MM-GBSA endpoint ($\Delta G_{\text{bind}} = (-18.25 \pm 0.40) \text{ kcal mol}^{-1}$; Table 5). The dissociation of two docked poses under explicit solvent illustrates why static docking scores alone are insufficient to predict binding persistence—the same limitation that motivates the divergence analysis in the Set-C pilot.

Docking-RRS and short-MD structural retention are non-equivalent quantities (central result)

The primary and secondary quantities are summarized in the Supporting Information (Tables S11, S12 and S22).

A 16-system pilot on PP-01 and PP-02 asked whether short trajectories and MM-GBSA would recover the docking-RRS pattern. All 16 trajectories met the geometric acceptance checks. For the eight mutants with a docking wild-type reference, docking predicted weaker mutant scores, while trajectory metrics more often indicated retained or tighter contacts (seven of eight comparisons disagreed in direction). This divergence is the principal calibration result: docking scores describe a static empirical scoring comparison, whereas the short MD metrics describe local geometric retention over the tested trajectory. Neither measurement is a direct estimate of affinity, residence time, target engagement, or resistance. MM-GBSA mutant/wild-type ratios exceeded 100 % for eight of twelve mutants; those ratios are endpoint comparisons, not free-energy measures of resilience. The pilot therefore does not reproduce docking-RRS and does not imply resistance or tighter binding.

As a bounded consistency check, we intersected the docking-RRS retention criterion (class-A, $\text{RRS} \geq 80 \%$) with pilot MD-RRS retention parity (mean minimum-distance ratio < 100) at pilot scope. Seven of twelve gate-eligible rows passed both gates: PP-01 satisfied the two-target gate (PfCRT via K76A, PfDHFR in four of four mutants), whereas PP-02 did not, because its PfDHFR states lack a docking wild-type anchor under the $5.0 \text{ kcal mol}^{-1}$ eligibility rule and only one target was gate-eligible. The single row above parity (PP-01

PfCRT K76T, ratio 102.7) lies within the $2.0 \text{ kcal mol}^{-1}$ endpoint noise floor and is read as retention, not gain (Table S19).

A single-ligand feasibility probe (PP-15, outside the Set-C cohort) extended the OpenFF protocol to wild-type PfDHFR and PfCRT; both systems completed 10 ns production trajectories with valid checkpoints, yielding MM-GBSA estimates of $(-29.52 \pm 0.33) \text{ kcal mol}^{-1}$ (PfDHFR) and $(-27.79 \pm 0.49) \text{ kcal mol}^{-1}$ (PfCRT); no RRS estimate is reported for this probe (Table S18).

Table 6: Set-C pilot MM-GBSA estimates (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots $\pm$ SEM; within-trajectory standard deviations are in the Supporting Information. MMG-RRS is the mutant/wild-type endpoint ratio ($\times 100$; wild type = 100). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 PfDHFR). Single-replicate endpoints should be read only as within-protocol contrasts.

Candidate	Target	State	ΔG_{bind} (kcal mol $^{-1}$)	SEM	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	—
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	—
PP-02	PfCRT	K76A	-24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	—
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	-30.16	2.06	100.0	—

External docking replication (sensitivity check)

Docking of 39 external ligands across the same eight PfDHFR/PfCRT states gave 312 finite scores. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means mutant and wild-type Vina scores were on average almost proportional, so this panel adds little mutant/wild-type contrast under the same protocol. It is a computational sensitivity check, not validation of the Set-C ranking or of experimental resistance (Table S19).

Discussion

The docking-RRS signal is reproducible but not a calibrated resistance predictor

The docking-derived RRS stratifies the 17-candidate Set-C cohort into five classes (A*–D) that separate predicted wild-type potency from predicted mutant retention. Under the declared protocol, this signal is operationally reproducible (seed dispersion ± 0.05 kcal mol⁻¹, 77.9% class stability under ± 1 kcal mol⁻¹ perturbation). However, the docking-derived triage is the baseline against which the diagnostic value of divergence is tested, not the paper’s principal contribution. The classes are combinatorially defined and not evidence of resistance circumvention; they can be aligned with regional resistance surveillance in a future decision analysis, but they do not justify clinical deployment on their own[?].

In the current epidemiological context, where the 2025 World Malaria Report documents declining artemisinin partner-drug efficacy alongside the spread of *Pfkelch13*-mediated partial resistance concentrated in the African Region[?], resistance-aware prioritization is a useful design criterion—though computational scores alone cannot establish whether a given compound is matched to the resistance states relevant in a parasite population.

The external docking panel (mean RRS ≈ 100) underscores the same boundary: mutant and wild-type Vina scores remained nearly proportional on an independent ligand set. The DEKOIS 2.0 PfDHFR benchmark (AUC 0.45, 95% CI 0.37–0.53) confirms that this protocol does not recover known actives above chance. A GNINA cross-scoring analysis reproduced the class-level picture (all 38 eligible ligands class A under both scoring layers; Table S19), indicating the retention-not-gain pattern is not a single-scoring-function artefact, though per-mutant rank transfer was weak or null (e.g. C59R -0.076). These observations caution against treating RRS as a calibrated resistance predictor outside the primary cohort.

The 87.5% divergence rate is a diagnostic signal for African anti-malarial NPs

The central question asked whether docking-RRS and short-MD structural retention are interchangeable estimates of mutant binding for African antimalarial NPs. The answer is no: seven of eight matched mutant–wild-type comparisons showed directional disagreement. This non-equivalence is not a failure of either method but a consequence of what each estimates—docking-RRS reflects a static score ratio between mutant and wild-type poses, while MD-RRS captures dynamic binding persistence under explicit solvent and protein fluctuations. Failure-mode analyses of docking confirm that cross-target score comparisons under non-comparable states and over-reading scores as affinities are recurrent pitfalls^{7 8}. The two quantities diverge when conformational selection or induced-fit effects alter the binding mode upon mutation, which is precisely the regime relevant to drug resistance.

The 87.5% divergence rate is high, and while general docking benchmarks document pose-reliability limitations, they rarely quantify docking–MD directional agreement for any specific chemical class⁷. Whether the elevated rate reflects properties specific to African natural products—rigid polycyclic frameworks, macrocyclic systems, and rare functional groups⁷—or a more general artefact of the short-MD protocol cannot be resolved within this study. No prior study has systematically quantified this divergence rate for African antimalarial NPs; the diagnostic signal it provides is the central finding of P2.

Short MD on the parent-study complexes shows both the value and the limit of structural follow-up when the compounds are not the ranked Set-C set. Two of four systems kept a bound ligand; the 164-labelled PfClpR and PfDHFR–201 systems did not. Only PfCRT–214 passed the checks needed for MM-GBSA, giving (-18.25 ± 0.40) kcal mol⁻¹—a system-specific endpoint that cannot calibrate Set-C scores^{7 8}. The Set-C pilot is the only trajectory comparison with docking-RRS for compounds on the prioritization list. Mutant MM-GBSA did not recover the docking-predicted weakening, so whether RRS tracks mutant-state free-energy differences remains unresolved. The directional mismatch observed in most mutant–

wild-type pairs persists even where the class-level retention gate was satisfied (PP-01).

For antimalarial lead prioritization, this means that docking-based RRS serves as a first-pass computational filter but is not a validated prediction of resistance resilience. MD stress tests, even short ones, provide a complementary but non-redundant signal that probes a different aspect of the binding event. Computational prioritization and structural scrutiny should therefore be reported as parallel ranking dimensions with non-overlapping failure modes, not as convergent evidence of resistance resilience. The most direct validation would be matched IC_{50} measurements on wild-type and mutant PfDHFR, PfCRT binding or transport assays at digestive-vacuole pH, and short growth-inhibition tests to determine whether multi-target computational signals translate to a measurable phenotype[?].

Chemical-space, network, and structural context

ACSI places candidates relative to approved-drug and African natural-product chemical space. The mean was 0.543, and two of 17 candidates (11.8%) exceeded 0.70. PNS is a network-weighted rank; on the two-target set it associated only weakly with RRS ($\rho = -0.2098$, adjusted $p = 1.0000$). At $n = 12$ the correlations mainly describe how the ranks line up in this cohort; the wide bootstrap intervals indicate that the three ranking dimensions capture distinct information. ACSI, PNS, and RRS are complementary ranking tools, not independent measures of biology, and their meaning depends on the chosen reference spaces, network, and targets^{???}.

Topological associations reported in a companion TDA analysis[?] used overlapping cohorts and are not independent validation; prospective testing in larger panels with experimental measurements is needed before topological descriptors can serve as design rules.

The PfCRT component of this picture is mechanistically plausible in light of long-timescale simulations of the chloroquine resistance transporter: 130 μ s of atomistic MD across CQ-sensitive and CQ-resistant isoforms established that Lys76 physically obstructs the drug-binding site while serving as the electrostatic anchor for peptide substrates, with R371I and

Q271E acting in concert with K76T to open drug access[?]. Our mutant-vs-wild-type end-point contrasts are consistent with this charged-cavity picture, in which perturbing residue 76 reshapes the local electrostatics that gate both ligand retention and substrate specificity. This consistency is contextual support, not an independent validation of the present docking scores.

Positioning within multi-target and resistance-aware approaches

Polypharmacological drug design has gained traction to overcome resistance in infectious diseases[?], including malaria[?]. Conventional multi-target docking ranks compounds by aggregated wild-type scores and does not treat mutant-state retention as a primary quantity. Network pharmacology often weights targets by centrality without comparing wild-type and resistance-associated structures under one scoring protocol[?]. Under the SDMT frame, the present class-A* readout combines exactly the two axes such a strategy requires—target breadth in the four-target wild-type matrix and per-target mutational retention in the RRS panel—and is therefore reported as an SDMT-style hypothesis, not as demonstrated multi-target engagement[?]. Free-energy methods such as FEP or TI report mutant binding differences with clearer physical content but remain costly for early triage. RRS is an explicit mutant/wild-type docking-score ratio that sits between those extremes, while ACSI, PNS, and short MD remain separate layers. None replaces IC₅₀ panels, transporter assays, or converged free-energy calculations; they are intended only to shortlist compounds for those experiments[?].

Limitations

Mutant receptors were homology models; all six met QMEAN, GMQE, Ramachandran, and RMSD thresholds, yet model quality is not experimental accuracy. MD was limited to 10 ns on four parent-study complexes and a single-replicate 16-system Set-C pilot; neither addresses long residence times or convergence. The JCIM MD-reporting guidelines

recommend at least three replicates per system[?]; no new replicate campaign was launched, and the pilot is therefore a secondary structural stress test, not a thermodynamic or kinetic measurement. The 87.5% divergence rate rests on eight matched comparisons from two candidates—the small denominator limits precision and prevents extrapolation to the full library. An independent PP-01 PfCRT wild-type run differed from the reference by 2.01 kcal mol⁻¹ (Table S17); single-replicate endpoints are sensitive to frame choice and periodic-boundary handling. MM-GBSA omits configurational entropy; only PfCRT-214 satisfied the bound-state and topology requirements. The four-target panel is not a full parasite targetome.

RRS class frequencies depend on the selection filters and unequal target coverage and should not be extrapolated to the full library. The cut-offs follow the Vina score distribution and have not been calibrated to biochemical resistance or clinical outcome (Table S6). PfCRT network centrality was imputed (Table S7) and PNS is sensitive to database coverage and hub bias[?]. Force fields differed between cohorts, so absolute MM-GBSA values are not compared across protocols. Cross-metric correlations rest on $n = 12$; the wide bootstrap intervals **indicate** that the three ranking dimensions capture distinct information. PfCRT was prepared at physiological pH rather than digestive-vacuole pH[?]. Five candidates lacked an eligible PfDHFR wild-type score and appear only in the PfCRT-only analysis.

Conclusion

Seven of eight matched mutant–wild-type comparisons showed directional disagreement between docking-RRS and short-MD structural retention (87.5% divergence rate), demonstrating that the two quantities are not interchangeable estimates of mutant binding for African antimalarial natural products. This elevated divergence rate suggests that African NPs—with their rigid polycyclic frameworks, macrocyclic systems, and rare functional groups—may be susceptible to static-pose artifacts, though whether the rate reflects properties spe-

cific to this chemical class or a more general artefact of the short-MD protocol cannot be resolved here. The divergence constitutes a diagnostic signal identifying candidates whose docking ranking requires experimental validation before further prioritization. Docking-RRS and MD-RRS should be reported as parallel ranking dimensions, not as convergent evidence of resistance resilience. The docking-derived five-class stratification is operationally reproducible under the declared protocol and serves as a hypothesis-generating prioritization. It does not constitute a validated triage procedure, nor does it demonstrate biological polypharmacology or experimental resistance circumvention. PP-01 **merits** experimental follow-up; PP-02 was not promoted by the pilot gate. Target-engagement assays, phenotypic tests, and prospective experiments in larger panels of African antimalarial NPs remain necessary before any biological claim can be advanced.

Authors' Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapel Njafa: Methodology, software, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. A versioned snapshot of this repository will additionally be archived on Zenodo to provide a permanent, citable DOI concurrent with publication. The large parent-study trajectories are not included in this article's data record; the conclusions drawn here are restricted to the

trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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Use of Artificial Intelligence

AI assistants were used to help with code and analysis scripts. All four Springer Nature AI-policy expectations were respected: (i) human accountability for every scientific claim remains with the listed authors; (ii) AI supported but did not replace scholarly judgement, editorial decision-making or the design of the comparison; (iii) the use of AI is disclosed here transparently; (iv) no confidential manuscript material, peer-review content or restricted data were shared with external AI services. All AI-assisted text and code were reviewed and edited by the authors. No generative AI tool is an author. Computational results, statistics, and quantitative claims were produced and checked by the authors.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and

Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and Paper 1 context tables (PDF).

TOC Graphic

